UNIVERSITÉ TOULOUSE CAPITOLE

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REPORT

Software Design

Assignment 1

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Chapter 1

Usecase Digrams

1.1 Indicate usecases and actors

From the description of the application, we have already defined the main actors of our application.

Actors	**Reason**
Customer	As mentioned in the assignment: "...customer starts a reservation by mentioning where they want to pick up the car..."
Agency Employee	As mentioned in the assingment: "Based on the chosen category, the agency employee will allocate a specific car to the customer."
Hotline Operator	As metioned in the assignment: "The 24/7 hotline will contact her back and organize with her the repairing actions"

Table 1.1: Actors in system

These actors with participate in several usecase:

Usecase	**Participation**
Make reservation	Customer
Pay for reservation	Customer
Cancel reservation	Customer
Pickup car	Customer, Agency Employee
Return car	Customer, Agency Employee
Manage incident	Customer, Hotline Operator
Report incident	Customer

Table 1.2: List of usecases and their participations

1.2 Usecases details

1.2.1 Make reservation

Name:	Make Reservation
Short description:	A customer reserve a car from the website of the company
Precondition:	
Postcondition:	A car reservation is made.
Error situations:	
System state in the event of an error:	
Actors:	Customer
Trigger:	Customer wanting to rent a car.
Standard process:	(1) Customer go to the website (2) Customer select the pickup location of the car (3) Customer select the pickup and return date of the car (4) Customer select the model of the car (5) Customer choose soome extra insurances (6) Customer choose extra options (7) Customer recieve extra

1.2.2 Pay for reservation

1.2.3 Cancel reservation

1.2.4 Pickup car

1.2.5 Manage incident

1.2.6 Report incident

1.3 Usecase diagram

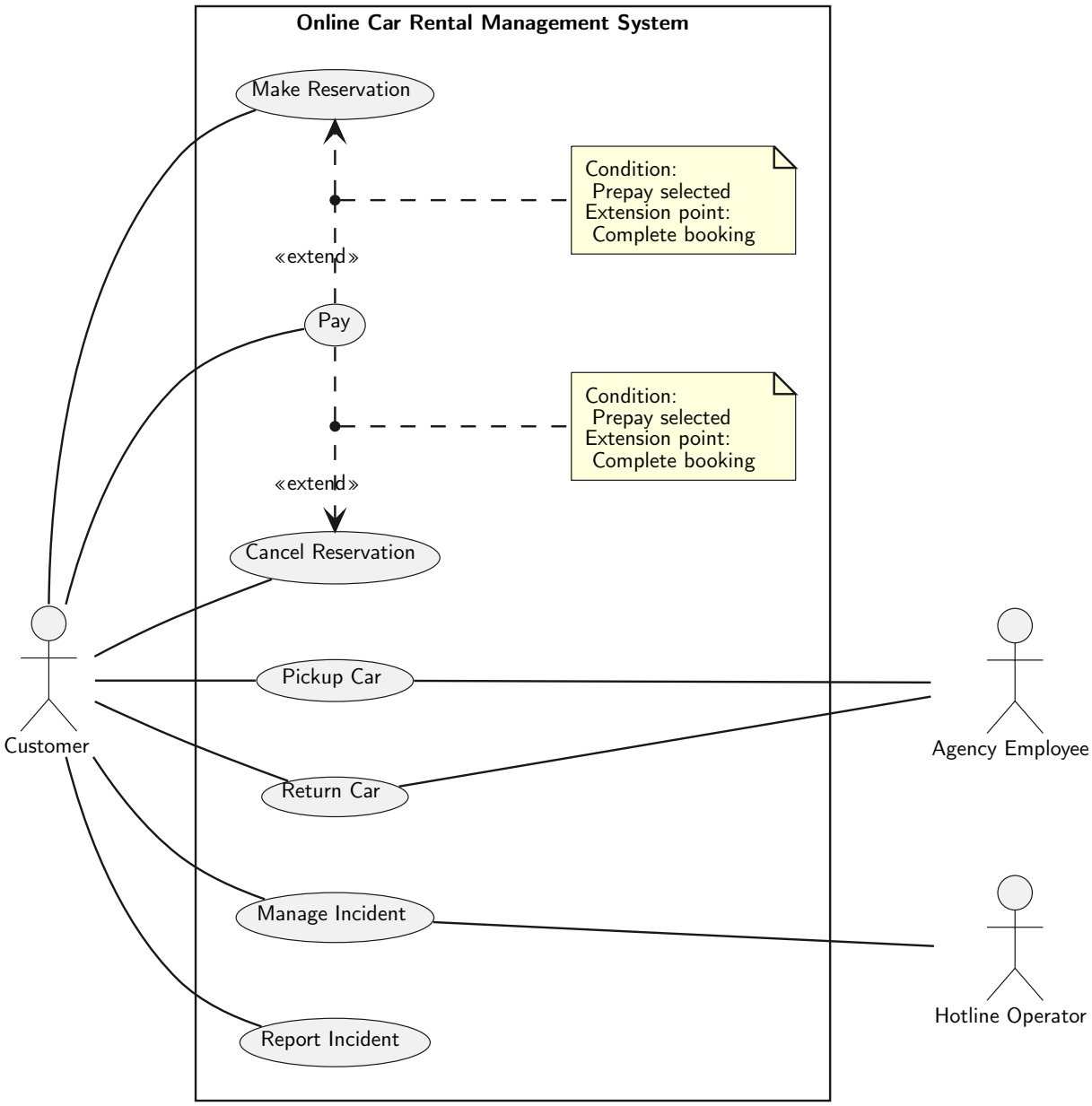


Figure 1.1: General usecase diagram

1.4 Detail of use cases associated to the action "pick up car"

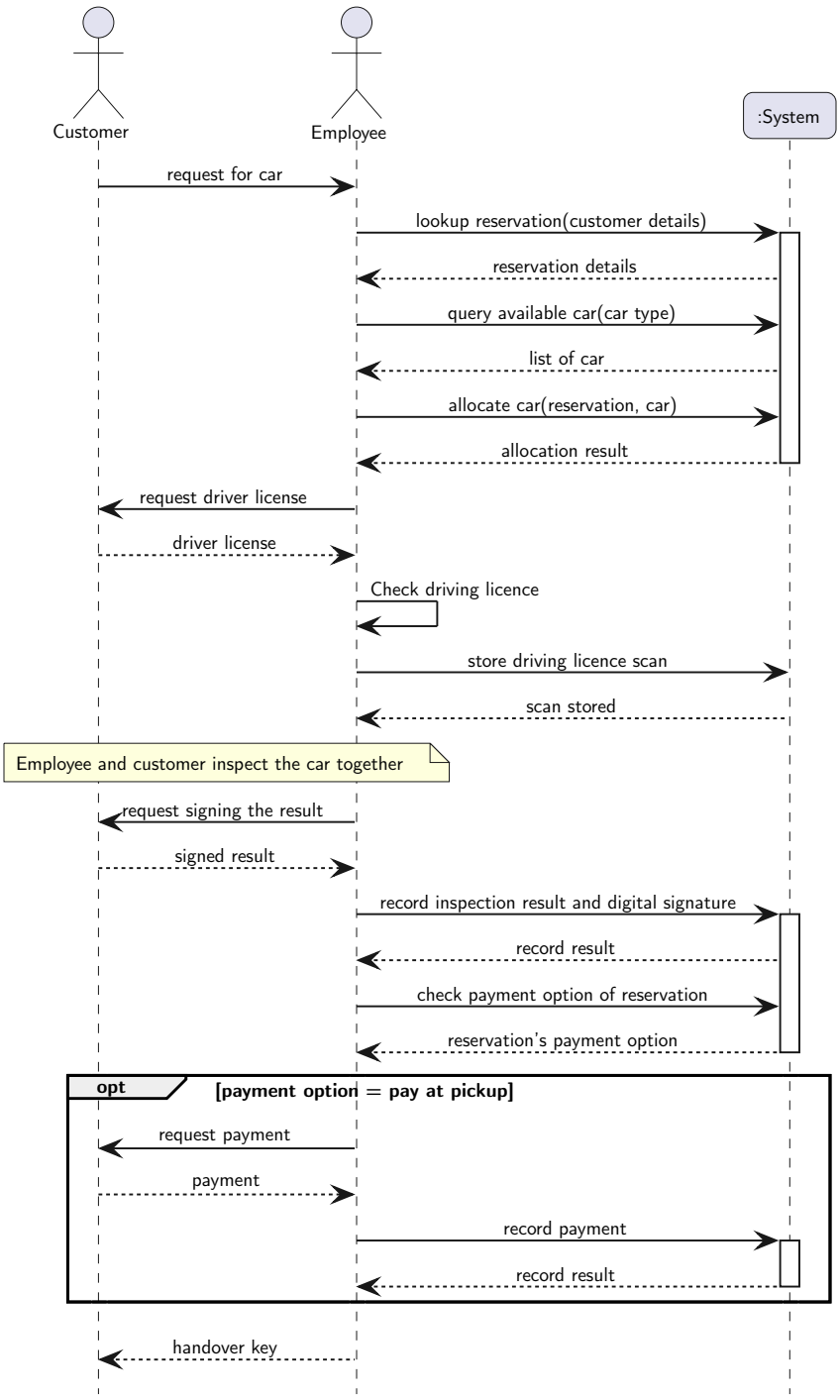

Customer
Employee
:System
request for car
lookup reservation(customer details)
reservation details
query available car(car type)
list of car
allocate car(reservation, car)
allocation result
request driver license
driver license
Check driving licence
store driving licence scan
scan stored
Employee and customer inspect the car together
request signing the result
signed result
record inspection result and digital signature
record result
check payment option of reservation
reservation's payment option
opt
[payment option = pay at pickup]
request payment
payment
record payment
record result
handover key

Figure 1.2:

Chapter 2

Class Diagram

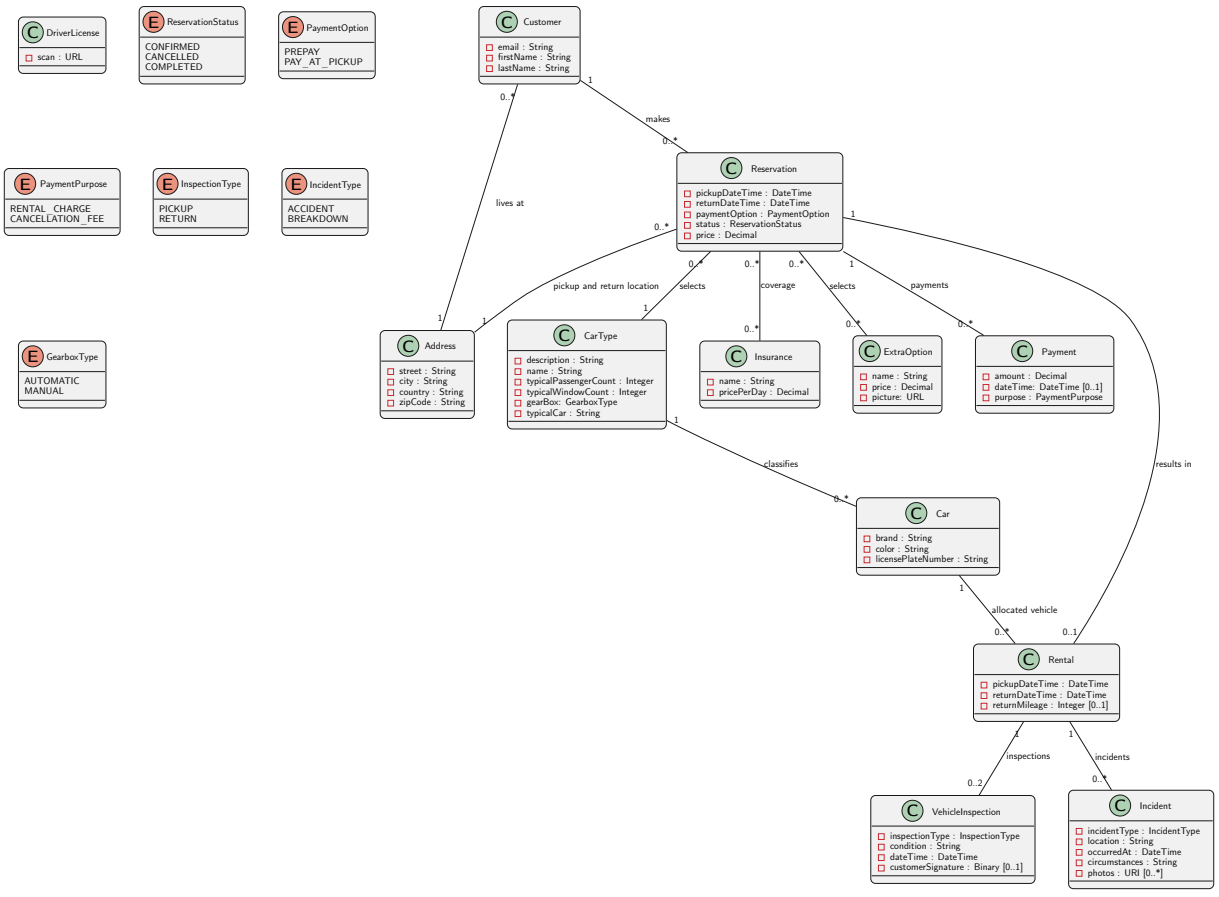


Figure 2.1: Class diagram

Chapter 3

Sequence Diagram

3.1 Reservation action

$$h_\theta(x) = \theta^T X \tag{3.1}$$

$$\text{ProbabilityOfClass}_1 = P(Y = 1|X) = \sigma(\theta^T X) = \frac{e^{\theta^T X}}{1 + e^{\theta^T X}} \tag{3.2}$$